

MTH102

Linear Algebra

2025–2026–I SEMESTER

PROBLEM SETS

24 September 2025

Chapter 1. Problem set 1

Exercise 1.1. Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$.
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$.

Exercise 1.2. What does it mean to say that A is not a subset of B ?

Exercise 1.3. Show that if A is a subset of B and B is a subset of C , then A is a subset of C .

Exercise 1.4. Show that no element n of $\mathbb{N}$ satisfies $n^4 - 5n^2 + 6 = 0$.

Exercise 1.5. What does it mean to say that two sets A, B are not equal?

Exercise 1.6. Consider the sets

$$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}.$$

Determine the elements common to these sets.

Exercise 1.7. Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.8. If A, B are sets, show that

$$A \subseteq A \cup B, B \subseteq A \cup B.$$

Exercise 1.9. If A, B are subsets of a set C , then $A \cup B$ is a subset of C .

Exercise 1.10. If A is a set, show that

$$A \cup A = A.$$

Exercise 1.11 (Commutative property). If A, B are sets, show that

$$A \cup B = B \cup A.$$

Exercise 1.12 (Associative property). If A, B, C are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

Exercise 1.13. Determine the intersection of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.14. If A, B are sets, show that

$$A \cap B \subseteq A, A \cap B \subseteq B.$$

Exercise 1.15. If A, B, C are sets satisfying

$$C \subseteq A, C \subseteq B,$$

then show that

$$C \subseteq A \cap B$$

holds.

Exercise 1.16. Identify the integers among

$$1, 2, 3, 4, \dots, 20$$

which are divisible by at least one of 2 and 5.

Exercise 1.17 (Inclusion-exclusion principle). Show that for finite subsets A, B, C of a set X ,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Exercise 1.18. Determine the number of integers among

$$1, 2, 3, 4, \dots, 100,$$

which are divisible by at least one of 6, 10, 15.

Exercise 1.19. If A is a set, show that

$$A \cap A = A.$$

Exercise 1.20 (Commutative property). If A, B are sets, show that

$$A \cap B = B \cap A.$$

Exercise 1.21 (Associative property). If A, B, C are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$

Exercise 1.22 (Distributive property). If A, B, C are sets, show that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Exercise 1.23. Show that

$$A^c = \{x \in X : x \notin A\},$$

where X denotes the underlying universal set.

Exercise 1.24. Determine the complement of $\{1, 2, 3, 4\}$ in $\{1, 3, 5, 7\}$, and the complement of $\{1, 3, 5, 7\}$ in $\{1, 2, 3, 4\}$.

Exercise 1.25. If A, B are subsets of a set X , show that

$$A^c \cap B = B \setminus A.$$

Exercise 1.26. If A is a subset of a set X , then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

Exercise 1.27. If A, B are subsets of a set X , then show that

$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

Exercise 1.28. Let A, B be subsets of a set X . Show that the following statements are equivalent.

$$\begin{aligned} A &\subseteq B, \\ A \cap B &= A, \\ A \cup B &= B, \\ B^c &\subseteq A^c. \end{aligned}$$

Exercise 1.29. If A, B, C are sets, show that

1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C),$
2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$

Exercise 1.30. Let A, B be subsets of a set X . Show that the sets

$$A \setminus B, B \setminus A$$

are disjoint.

Exercise 1.31. Determine the symmetric difference of $\{1, 2, 3, 4\}$ and $\{1, 3, 5, 7\}$.

Exercise 1.32. If A, B are sets, then show that

$$\begin{aligned} A \Delta B &= (A \setminus B) \cup (B \setminus A), \\ A \Delta B &= B \Delta A, \\ A \cup B &= (A \Delta B) \cup (A \cap B), \\ (A \Delta B) \cap (A \cap B) &= \emptyset. \end{aligned}$$

Exercise 1.33. Do there exist three finite sets A, B, C satisfying

$$|A \Delta B| = 1, |B \Delta C| = 2, |C \Delta A| = 4?$$

Exercise 1.34. If A, B are finite sets, then show that

$$|A \times B| = |A| \times |B|,$$

where for a set X , its number of elements is denoted by $|X|$.

Exercise 1.35. Determine the cartesian product of $[1, 2]$ and $[3, 4] \cup [5, 6]$.

Exercise 1.36. Identify the set

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\}.$$

Exercise 1.37. Let A, B be subsets of $\mathbb{R}$ defined by

$$A = \{x \in \mathbb{R} : x^2 \geq 0\},$$

$$B = \{x \in \mathbb{R} : x^3 \geq 0\}.$$

Determine the sets $A \cup B, A \cap B, A \setminus B, B \setminus A$.

Exercise 1.38. For a positive integer k , let A_k denote the set of integral multiples of k , that is,

$$A_k := \{r \in \mathbb{Z} : r = k\ell \text{ for some } \ell \in \mathbb{Z}\}.$$

Let m, n be positive integers. For each of the following statements, determine the equivalent conditions on the integers m, n .

1. $A_m \subseteq A_n$
2. $A_m \subsetneq A_n$
3. $A_m \not\subseteq A_n$
4. $A_m = A_n$
5. $A_m \cap A_n = \emptyset$
6. $A_m \setminus A_n \neq \emptyset$
7. $A_m \setminus A_n \neq \emptyset$ or $A_n \setminus A_m \neq \emptyset$
8. $A_m \setminus A_n \neq \emptyset$ and $A_n \setminus A_m \neq \emptyset$

Exercise 1.39. Show that $\mathbb{R}$ is a subset of $\mathbb{C}$.

Exercise 1.40. If z, w are complex numbers, then show that

1. $\overline{z + w} = \overline{z} + \overline{w}$,
2. $\overline{z \cdot w} = \overline{z} \cdot \overline{w}$,
3. $|z| \geq 0$, and $|z| = 0$ if and only if $z = 0$,
4. $|z \cdot w| = |z| |w|$,
5. $|z + w| \leq |z| + |w|$ (triangle inequality).

Exercise 1.41. If z is a complex number, then show that

$$z \cdot \overline{z} = |z|^2.$$

Exercise 1.42. For any real number θ , show that

$$|\cos \theta + i \sin \theta| = 1.$$

Exercise 1.43. If z is a complex number satisfying $|z| = 1$, then show that there exists a real number θ such that

$$z = \cos \theta + i \sin \theta.$$

Exercise 1.44. If z is a nonzero complex number, then show that

$$z = |z| (\cos \theta + i \sin \theta)$$

for some real number θ .

Chapter 2. Induction principles

Exercise 2.1. If A is a nonempty subset of $\mathbb{N}$, and a_1, a_2 are least elements of A , then show that $a_1 = a_2$.

Exercise 2.2. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.3. Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.4. Show that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n \in \mathbb{N}$.

Exercise 2.5. Show that

$$3 + 11 + \cdots + (8n-5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Exercise 2.6. Show that

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}$$

for all $n \in \mathbb{N}$.

Exercise 2.7. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Exercise 2.8. Show that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

hold for any positive integer n .

Exercise 2.9. Show that

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

for all $n \in \mathbb{N}$.

Exercise 2.10. Show that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Exercise 2.11. Show that $5^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Exercise 2.12. Show that $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.

Exercise 2.13. Show that $6^n - 5n - 1$ is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 2.14. Show that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Exercise 2.15. Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Exercise 2.16. Show that $3^n \geq n^2$ for all $n \in \mathbb{N}$.

Exercise 2.17. Show that $n! > 2^n$ for all $n \in \mathbb{N}$ satisfying $n \geq 4$.

Here are a few exercises where strong induction is useful.

Exercise 2.18. Show that every positive integer greater than 1 is a prime or is a product of prime numbers.

Exercise 2.19. Let the integers $x_1, x_2, \dots$ be defined by

$$x_1 := 1,$$

$$x_2 := 2,$$

$$x_{n+2} := \frac{1}{2}(x_n + x_{n+1})$$

for all $n \in \mathbb{N}$. Show that $1 \leq x_n \leq 2$ for all $n \in \mathbb{N}$.

Exercise 2.20. The Fibonacci sequence $F_0, F_1, F_2, \dots$ is defined by

$$F_0 := 0,$$

$$F_1 := 1,$$

$$F_2 := 1,$$

$$F_{n+2} := F_n + F_{n+1}$$

for all $n \in \mathbb{N}$. Show that

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right)$$

for all $n \in \mathbb{N} \cup \{0\}$.

Exercise 2.21. Show that

$$F_0^2 + F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$$

for all $n \in \mathbb{N} \cup \{0\}$.

Chapter 3. Matrices

Exercise 3.1. If A, B, C are matrices of the same size, show that

$$A + (B + C) = (A + B) + C.$$

Exercise 3.2. If A, B are matrices of the same size, show that

$$A + B = B + A.$$

Exercise 3.3. If A is an $m \times n$ matrix, show that

$$A + 0_{m \times n} = 0_{m \times n} + A = A$$

holds, where $0_{m \times n}$ denotes the $m \times n$ zero matrix, that is, the $m \times n$ matrix whose entries are all zero.

Exercise 3.4. Show that if A is a matrix and c, d are real numbers, then

$$(c + d)A = cA + dA$$

and

$$c(dA) = (cd)A.$$

Exercise 3.5. If A is a matrix, what is $(A^T)^T$?

Exercise 3.6. Show that if A, B are matrices of the same size, then

$$(A + B)^T = A^T + B^T.$$

Exercise 3.7. Show that if A denotes a matrix, then

$$(A^T)^T = A.$$

Exercise 3.8. Show that if A, B, C are matrices such that the products $A(BC)$ and $(AB)C$ are defined, then

$$A(BC) = (AB)C.$$

Exercise 3.9. Provide examples to show that in general, matrix multiplication is not commutative, that is, $AB \neq BA$ for some matrices A, B .

Exercise 3.10 ().** Show that for any $n \in \mathbb{N}$ with $n \geq 2$, there are $n \times n$ matrices A, B such that

$$AB \neq BA.$$

? Question

Can induction be used for the above exercise?

Exercise 3.11. If A is an $n \times n$ matrix, show that

$$AI_n = I_n A = A$$

holds, where I_n denotes the $n \times n$ diagonal matrix, with all diagonal entries equal to 1, that is,

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

Exercise 3.12. If A is an $m \times n$ matrix, show that

$$I_m A = A, A I_n = A$$

where I_m (resp. I_n) denote the $m \times m$ (resp. $n \times n$) identity matrix.

Exercise 3.13. If A is an $m \times n$ matrix with entries in $\mathbb{R}$ and c is a real number, then

$$cA = (cI_n)A = A(cI_m),$$

where I_n (resp. I_m) denotes the $n \times n$ (resp. $m \times m$) identity matrix.

Exercise 3.14. Show that if A, B are matrices, then

$$(AB)^T = B^T A^T.$$

Exercise 3.15. Show that if

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

then

$$A^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}$$

for all $n \in \mathbb{N}$ with $n \geq 1$, where F_n denotes the n -th Fibonacci number.

Exercise 3.16. Compute

$$\begin{pmatrix} 0 & \dots & 0 & 1 \\ 0 & \dots & 1 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 1 & \dots & 0 & 0 \end{pmatrix}^2.$$

Exercise 3.17. Show that if A, B, C are matrices such that the sum $B + C$, and the products $AB, AC, A(B + C)$ are defined, then

$$A(B + C) = AB + AC.$$

Exercise 3.18. Show that if A, B, C are matrices such that the sum $B + C$, and the products $AB, AC, A(B + C)$ are defined, then

$$(A + B)C = AC + BC.$$

Exercise 3.19. If A is an $m \times n$ matrix, then show that for all $i = 1, 2, \dots, n$, the product Ae_i is equal to the i -th column of A , where e_i denotes the i -th standard basis vector of $\mathbb{R}^n$, that is,

$$e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

with 1 in the i -th position, and 0's elsewhere.

Exercise 3.20. Let A, B be invertible $n \times n$ matrices. Show that AB is invertible, and that

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Compare the above exercise with [Exercise 3.14](#).

Exercise 3.21. If A, B are 2×2 matrices, then show that

$$\det(AB) = \det(A) \det(B).$$

Exercise 3.22. Solve the system of equations

$$\begin{aligned} 5x + 2y &= 11, \\ 3x + 4y &= 8. \end{aligned}$$

Exercise 3.23. Solve the system of equations

$$\begin{aligned} 12x - 25y &= -47, \\ -7x + 30y &= 51. \end{aligned}$$

Exercise 3.24. Solve the system of equations

$$\begin{aligned} 2x + 3y + z &= 1, \\ 4x + y + 2z &= 2, \\ 3x + 2y + 3z &= 3. \end{aligned}$$

Exercise 3.25. Let A, B be 2×2 matrices. Show that if AB is invertible, then both A and B are invertible.

Exercise 3.26. Let A be an $n \times n$ matrix. If P is an invertible $n \times n$ matrix, then show that

$$(PAP^{-1})^k = PA^kP^{-1}$$

holds for any positive integer k .

Exercise 3.27. Let A be a 2×2 matrix. Let V be the set of solutions to the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

that is,

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V , and the element cu also lies in V .

Exercise 3.28. Let A be a 2×2 matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is¹,

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} s \\ t \end{pmatrix} \text{ for some } s, t \in \mathbb{R} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V , and the element cu also lies in V .

Exercise 3.29. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations $Au = 0$,

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

Exercise 3.30. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

Exercise 3.31. Show that for any $n \geq 2$, there are infinitely many $n \times n$ orthogonal matrices.

Exercise 3.32. Let A be an orthogonal matrix. Show that A is invertible and that

$$A^{-1} = A^T.$$

Exercise 3.33. Let A, B be orthogonal matrices of the same size. Show that the matrix product AB is also an orthogonal matrix.

Exercise 3.34. Let A, B be square matrices with complex entries. Show that

$$\overline{AB} = \overline{A} \overline{B}.$$

Exercise 3.35. Let A be a square matrix with complex entries. Show that

$$\overline{A^T} = (\overline{A})^T.$$

Exercise 3.36. Let A, B be square matrices of the same size with complex entries. Show that

$$(A + B)^* = A^* + B^*.$$

Exercise 3.37. Let A be a matrix with complex entries. Show that

$$(A^*)^* = A.$$

Exercise 3.38. Let A, B be matrices with complex entries. Show that

¹Observe that

$$A \begin{pmatrix} s \\ t \end{pmatrix} = sC_1 + tC_2,$$

where C_1, C_2 denote the first and second columns of A respectively.

$$(AB)^* = B^* A^*.$$

Exercise 3.39. Let A be an $n \times n$ hermitian matrix. Show that all the diagonal entries of A are real numbers. More generally, show that for any $1 \leq i, j \leq n$, the (i, j) -entry and the (j, i) -entry of A are complex conjugates of each other.

Exercise 3.40. Let A, B be hermitian matrices of the same size. Show that the matrix $A + B$ is also a hermitian matrix. Is the matrix AB also a hermitian matrix?

Exercise 3.41. State and prove an analogue of [Exercise 3.39](#) for skew-hermitian matrices.

Exercise 3.42. Let A, B be skew-hermitian matrices of the same size. Show that the matrix $A + B$ is also a skew-hermitian matrix. Is the matrix AB also a skew-hermitian matrix?

Exercise 3.43. Let A be a square matrix with complex entries. Show that if A is hermitian and skew-hermitian, then A is the zero matrix.

Exercise 3.44. Let A be a unitary matrix. Show that A is invertible and that

$$A^{-1} = A^*.$$

Exercise 3.45. Let A, B be unitary matrices of the same size. Show that the matrix product AB is also a unitary matrix.

Exercise 3.46. Show that any orthogonal matrix is a unitary matrix. Is the converse true?

Exercise 3.47. Determine whether the following matrices are unitary, hermitian, both, or neither.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Exercise 3.48. Let z, w be complex numbers such that $|z|^2 + |w|^2 = 1$. Show that

$$\begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix}$$

is a unitary matrix.

Exercise 3.49. Let A be a diagonal matrix with complex entries. Show that A is a unitary matrix if and only if all the diagonal entries of A have absolute value equal to 1.

Exercise 3.50. Show that any unitary matrix is a normal matrix. Is the converse true?

Exercise 3.51. Show that any hermitian matrix is a normal matrix. Is the converse true?

Exercise 3.52. Show that any skew-hermitian matrix is a normal matrix. Is the converse true?

Exercise 3.53. Let e_{ij}, e_{kl} be matrix units of the same size. Show that $e_{ij}e_{kl}$ is equal to the matrix unit e_{il} if $j = k$, and is equal to the zero matrix if $j \neq k$, that is,

$$e_{ij}e_{kl} = \begin{cases} e_{il} & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

Exercise 3.54. Solve the system of linear equations in five variables x_1, x_2, x_3, x_4, x_5 :

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 + x_5 &= 7, \\ 2x_1 + 3x_2 + x_3 + 5x_4 + 2x_5 &= 14, \\ -x_1 + 4x_2 - 2x_3 - 3x_4 + x_5 &= -10, \end{aligned} \tag{1}$$